

Co-Pilot for Lawyers

An AI assistant that reads legal documents, extracts data, flags rule deviations, and answers questions with auditable source citations.
Group 88: Konstantin Unterweger, Vincent A. Balau, David Fellner, Florian Ebner | Applied Generative AI and LLM-based Systems · 2026

Abstract

Lawyers **spend a disproportionate amount of time reading** contracts and regulatory documents.

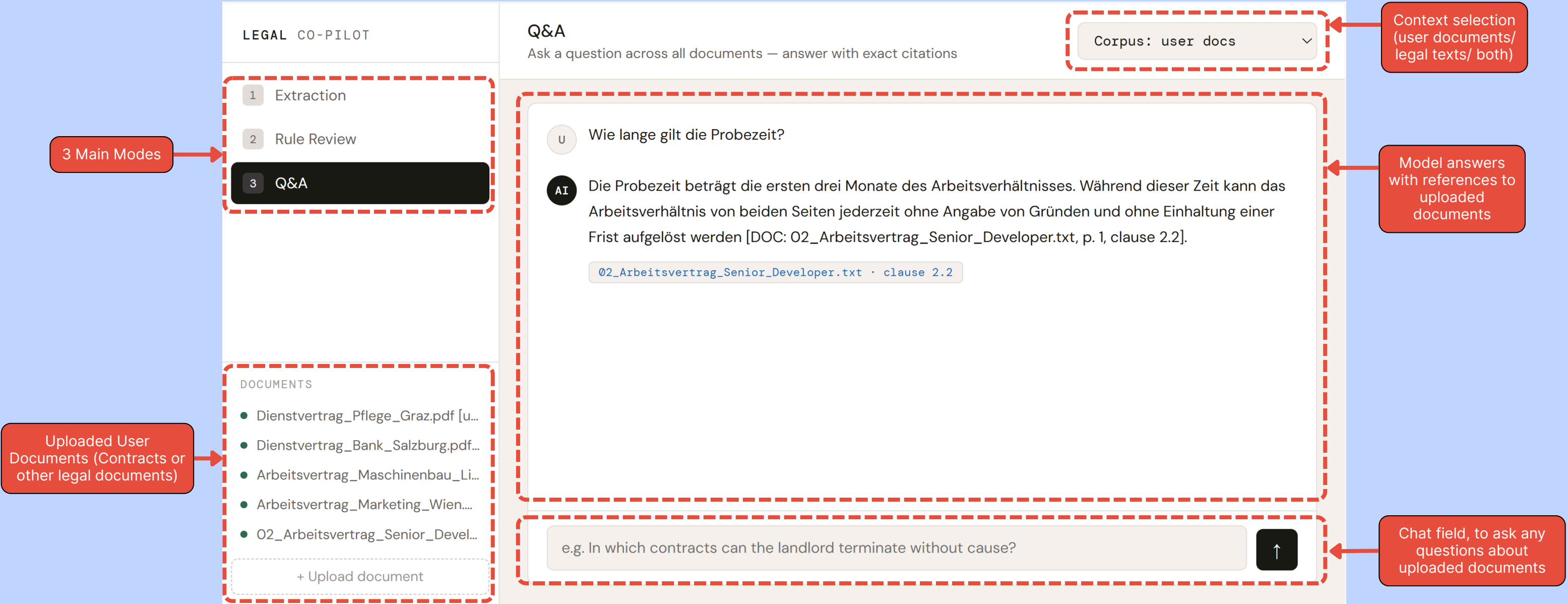
Our system addresses three core tasks: **Extraction** (pulling structured data from unstructured text), **Rule-based Review** (flagging deviations from predefined rules), and **Cross-document Q&A** (answering questions across multiple documents).

A key design principle is **source grounding** — every answer includes precise citations to the originating document passages, ensuring auditability and trust.

Initial Situation & Problem

Three recurring pain points for legal professionals:

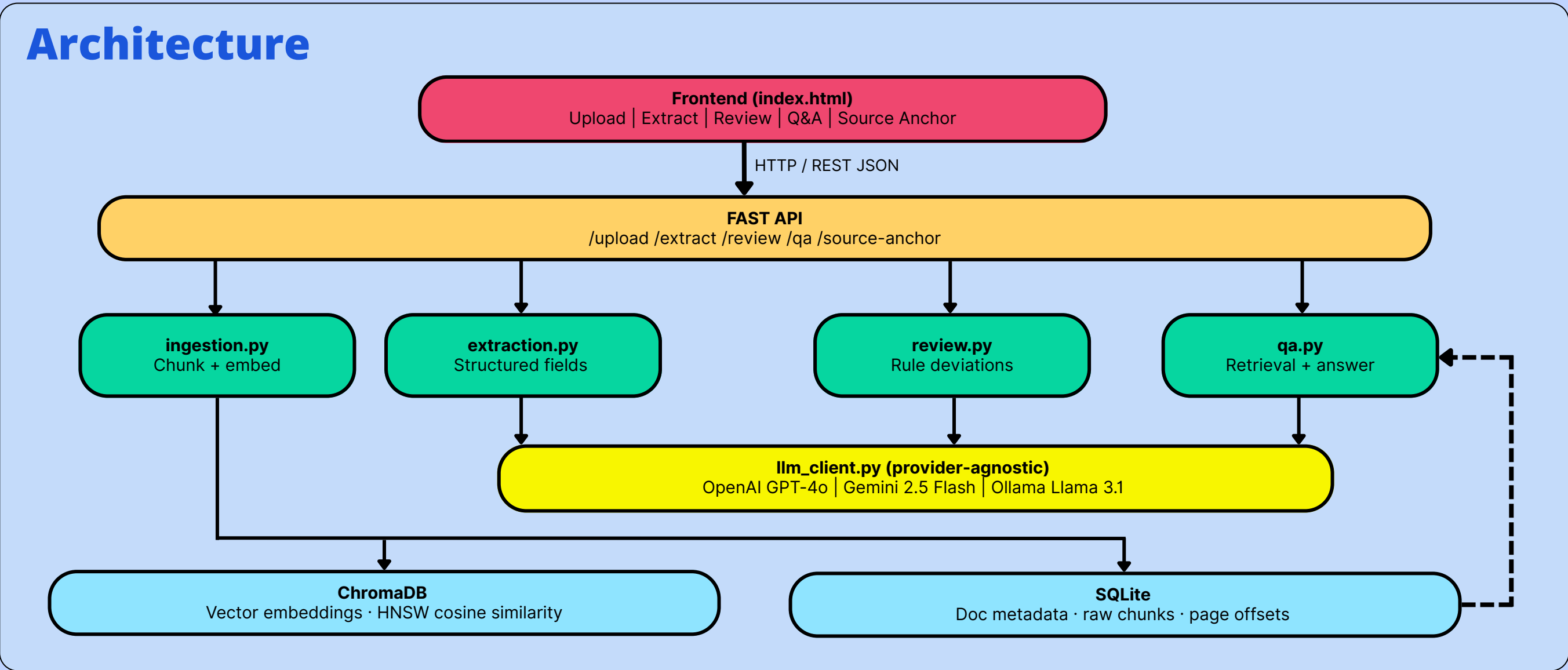
- 1. **Multi-document extraction** — manually pulling key terms, dates, and clauses from dozens of contracts.
- 2. **Risk review** — checking compliance against internal policies and regulatory frameworks.
- 3. **Cross-document research** — finding answers that span multiple documents without missing context.



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Architecture



Key Design Decisions

- 1 Dual Storage: Vector + Relational**
ChromaDB serves semantic search (HNSW cosine).
SQLite is the authoritative store for metadata, raw chunk text, page numbers and char offsets.
- 2 Chunk Based Retrieval.**
Chunks size: ~1500-char windows with 200-char overlap.
Each chunk carries its page number and char offsets.
Only semantic relevant chunks are given to the LLM
- 3 Switchable LLM Provider**
OpenAI, Gemini, and Ollama (local) can be switched via LLM_PROVIDER without modifying the code (llm_client.py:23). This enables on-premises operation with Ollama for law firms with data protection concerns. Trade-off: the lowest common denominator determines the architecture—features such as OpenAI Structured Outputs or Gemini-specific grounding APIs are not utilized.
- 4 RAG Instead of Fine-Tuning**
Each query retrieves chunks fresh from the database and passes them to the LLM as context. This means new documents are immediately available without any training effort. Fine-tuning would provide better quality for technical language, but it would be expensive and would have to be repeated for every new document type.

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LEGAL CO-PILOT

1 Extraction

2 Rule Review

3 Q&A

Mode selection

Document selection

DOCUMENTS

Dienstvertrag_Pflege_Graz.pdf [u...

Dienstvertrag_Bank_Salzburg.pdf...

Arbeitsvertrag_Maschinenbau_Li...

Arbeitsvertrag_Marketing_Wien...

O2_Arbeitsvertrag_Senior_Devel...

+ Upload document

Extraction

Define fields — get a structured table with exact source links

corpus selection dropdown menu

Corpus: user docs

Run ↗

Button to run the extraction feature

Results — 5 documents × 5 fields

← Edit

Button to edit the fields for extraction

Document	Berufsbezeichnung	Startdatum	Monatsgehalt	Bonusregelung	Arbeitszeit (Stunden pro Woche)
Dienstvertrag_Pfleg...	diplomierte Gesundheits- und Krankenpflegerin (DGKP) clause 3.1 found	1. Juni 2025 § 2.1 found	EUR 3.420,-- § 5.1 found	— p. 2 not_specified	37 § 4.1 found
Dienstvertrag_Bank_...	Head of Credit Risk Management § 3.1 found	1. September 2025 § 2.1 found	10.142,86 Page 2, clause 5.1 found	variable Vergütungssystem Page 2, paragraph 5.2 found	38,5 § 4 Abs. 1 found
Arbeitsvertrag_Masc...	Senior Konstrukteur § 3.1 found	1. Mai 2025 § 2.1 found	EUR 4.850,-- § 5.1 found	Projektprämie 5.3 found	38,5 § 4.1 found
Arbeitsvertrag_Mark...	Junior Content & Social Media Manager § 3.1 found	1. März 2025 § 2.1 found	EUR 2.650,-- § 5.1 found	— p. 2 not_specified	38,5 Stunden § 4.1 found
O2_Arbeitsvertrag_S...	Senior Software Developer § 3.1 found	1. Jänner 2025 § 2.1 found	EUR 6.800,-- § 5.1 found	15% des Jahresbruttogehalts	38,5 § 4 Abs. 1 found

specified and successfully identified field in the legal text

Result row for each Document

field not specified in legal document

✓ What Worked

- Dual storage (vector + relational) enabled fast semantic search with full audit trails.
- Provider-agnostic LLM layer allowed rapid experimentation across models.
- Paragraph-aware chunking improved retrieval relevance significantly.

✗ What Didn't

- Rule translation from natural language to machine-checkable constraints remains brittle.
- Absence detection (proving something is NOT in a document) is unreliable.
- Chunking across tables and nested clauses loses structural context.

→ Next Iteration

- Build a dataset for systematic evaluation of retrieval and extraction.
- Introduce a versioned rule playbook for traceable compliance checks.
- Adopt structured PDF parsing (layout-aware) to preserve tables and hierarchies.